CITRUS: Candidate Inference, Temporal Decoding, and Reliability-Gated Reporting for Wearable Devices

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Abstract

Wearable photoplethysmography (PPG) enables continuous heart rate (HR) monitoring on consumer devices, but motion artifacts introduce frequency-domain ambiguity: during exercise, the dominant spectral component may correspond to limb movement rather than the true pulse. This paper presents CITRUS, a two-layer system that decomposes motion-robust HR monitoring into *estimation* (candidate selection + causal Viterbi decoding) and *reporting* (reliability scoring + accept/hold/reject policy). The platform benchmark contains 54 participants and 49.76 h of recordings against clinical references; the motion-focused ring-PPG evaluation uses 31 subjects. On this subset, the estimation layer reduces motion-window Mean Absolute Error (MAE) from 14.39 to 10.91 beats per minute (BPM). We additionally develop an ESP32-S3 real-time prototype for live PPG acquisition, on-device filtering, beat-event extraction, and browser visualization. We deploy the feature-vector CITRUS inference core—candidate scoring, causal decoding, and reliability estimation—on the ESP32-S3, averaging 156.429 ms per window with 145,280 B model storage.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**; • **Applied computing** → *Health informatics*.

Keywords

wearable sensing, photoplethysmography, heart rate, motion artifacts, ESP32, ring

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1 Introduction

Wearable devices—smartwatches, rings, earbuds—increasingly embed photoplethysmography (PPG) sensors for continuous heart rate (HR) monitoring [5, 14, 16]. A PPG sensor illuminates the skin and records intensity changes caused by blood-volume variation [1]. Under stationary conditions, ring and wrist PPG can track HR close to clinical-reference accuracy. However, motion changes sensor–skin contact, local pressure, and optical coupling, introducing artifacts whose frequencies overlap the cardiac band [7, 17]. During exercise, the largest spectral peak may correspond to step frequency or a harmonic rather than the true pulse. Motion-corrupted PPG is therefore not only a denoising problem but also an *ambiguity* problem: a single analysis window can contain multiple plausible HR values.

Most HR estimation systems report a value for every window [13]. This metric measures estimator accuracy, but it does not reflect deployment reality. On a wearable device, a wrong value displayed with high confidence is worse than a short gap or a held previous value. A deployable system must therefore answer two questions: (i) *estimation*—given a corrupted window, which HR value is most likely correct? (ii) *reporting*—should the device emit the current estimate, hold the previous trusted value, or reject the window? These two questions are coupled but distinct: a better estimator does not eliminate all unreliable windows, and severe motion can make the cardiac component non-identifiable from the available signal alone.

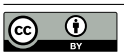
This paper presents CITRUS, a two-layer system that explicitly separates estimation from reporting. The **estimation layer** converts each PPG window into approximately 140 candidate HR values using multiple signal-processing estimators, scores candidates with a learned ranker, and applies causal Viterbi decoding with a physiological transition penalty. The **reporting layer** estimates per-window reliability and applies a learned finite-state policy that decides whether to accept the current estimate, hold the previous trusted value, or reject the window. This separation makes the coverage–error trade-off explicit: users and downstream applications can choose between high-coverage monitoring and high-confidence spot-checking.

2 Related Work

2.1 PPG Heart-Rate Estimation under Motion

Ring-worn PPG is increasingly studied for continuous monitoring, including atrial-fibrillation screening [8], oxygen saturation [11], and cuffless blood pressure. A major obstacle across wearable form factors is motion. Mechanical coupling between the optical path and limb movement injects artifacts whose frequencies overlap the cardiac band [17]. Mitigation strategies include adaptive filtering with accelerometer references [9], signal decomposition with sparse

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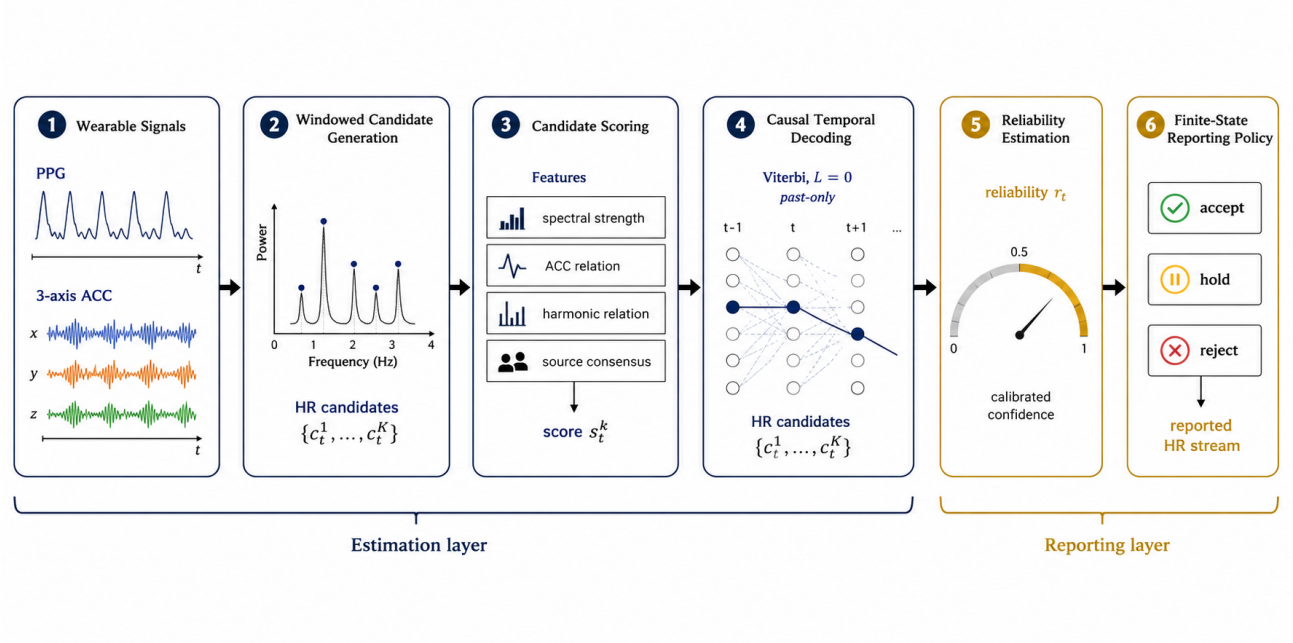


Figure 1: CITRUS pipeline: the estimation layer converts each PPG window into candidates, scores them, and applies causal Viterbi decoding; the reporting layer estimates reliability and applies an accept/hold/reject policy.

spectral reconstruction [17], and deep learning on large-scale activity datasets [13]. Classical spectral trackers rely on hand-designed peak-selection rules, whereas end-to-end regression models provide limited visibility into which spectral hypothesis caused an error. This work adopts candidate selection as the estimation formulation: multiple digital signal processing (DSP) estimators generate HR hypotheses, and a learned scorer ranks them using source, spectral, consensus, and temporal features.

2.2 Temporal Decoding and Selective Prediction

Heart rate is temporally continuous and generally changes gradually across adjacent analysis windows. Per-window candidate scoring may incorporate local information from the previous estimate, but it does not enforce path-level temporal consistency across multiple windows. Selecting one value per window from a candidate set under a smoothness constraint can be formulated using Viterbi dynamic programming [12, 15]. This paper uses an online causal recursion that updates the path cost using the current window and an accumulated state derived only from past windows, preserving real-time deployability.

Selective prediction studies when a model should return an output [4, 6]. In wearable HR estimation, a reliability threshold should not be assumed device-independent: PPG quality, motion patterns, and candidate distributions vary across devices and cohorts. This paper treats reliability estimation as a calibration problem tied to the sensing setting, rather than using a fixed universal threshold.

2.3 On-Device Inference

Wearables operate on microcontroller-class hardware with tight power budgets. The tiny machine learning (TinyML) literature targets this regime through compact architectures and hardware–software co-design [3, 10]. Prior work has demonstrated on-sensor beat detection at microwatt power [2]. This paper reports measured ESP32-S3 latency, storage, and memory and a datasheet-based estimate of inference energy for the feature-vector inference core while treating raw-signal integration as a staged migration step.

3 System Design

3.1 Estimation Layer

Candidate generation. Each 30-second ring-PPG window (25 Hz; infrared (IR) and red channels) is processed by three estimator families: Welch spectral peaks, autocorrelation peaks, and peak-interval estimates. Harmonic variants (0.5×, 1×, 2×) are added, and values outside [35, 220] BPM are removed. The resulting candidate set C_t contains approximately 140 valid candidates per window (median 140, range 57–160). The accelerometer (ACC) is used before candidate extraction to select motion-quiet segments for spectral estimation; it is not a candidate source.

Candidate scoring. Each candidate $c_{t,k}$ is represented by features $\phi_{t,k}$: source method, source channel, spectral power, consensus among nearby candidates, harmonic status, and distance to the previous window's decoded value. For the primary 31-subject evaluation, a small multilayer perceptron (MLP) maps these features to candidate-ranking logits, with soft targets centered on the reference HR. The MLP scorer has 28k parameters. A raw-waveform convolutional neural network (CNN) encoder was also tested on this subset but provided no measurable benefit: removing it improves overall MAE by 0.01 BPM while reducing the parameter count from 85k to 28k. For the sustained-motion stress test, we instead use a

histogram-based gradient-boosting regressor (HGB) to score the structured candidate features. This HGB candidate scorer is also used in the ESP32-S3 feature-vector deployment benchmark.

Causal Viterbi decoding. Candidate scoring alone can still produce physiologically implausible (e.g., 72 BPM → 144 BPM → 70 BPM). We therefore apply an online Viterbi recursion that updates the minimum cumulative cost whenever a new window arrives. Let $D_t(k)$ denote the minimum cost of a candidate path ending at candidate $c_{t,k}$ in window t:

$$\begin{aligned} D_1(k) &= E_1(k), \\ D_t(k) &= E_t(k) + \min_j \left[D_{t-1}(j) + \Omega_t(j,k) \right], \\ \hat{k}_t &= \operatorname*{argmin}_k D_t(k), \qquad \hat{h}_t = c_{t,\hat{k}_t}. \end{aligned} \tag{1}$$

Here, $E_t(k)$ is the candidate cost derived from the candidate scorer, and $\Omega_t(j,k)$ is a quadratic transition penalty between adjacent HR candidates. Because the update depends only on the current window and the accumulated costs from window $t-1$, the estimate $\hat{h}_t$ can be produced without observing future windows or performing backward smoothing.

In a controlled decoder ablation conducted using a fixed data split and random seed, the causal and full-sequence variants achieve 10.66 and 10.88 BPM motion MAE, respectively. These single-run ablation results are used only to compare the two decoding strategies. In the primary evaluation averaged over five random seeds, the causal pipeline achieves 10.91 ± 0.73 BPM motion MAE, compared with 12.94 BPM before temporal decoding.

3.2 Reporting Layer

Reliability scoring. The estimation layer always produces a value, but under heavy motion all candidates may be unreliable. An HGB model estimates per-window reliability using PPG quality metrics, inter-estimator disagreement, candidate dispersion, and motion intensity (44–57 features). The target is a continuous reliability score $r_t = \exp(-|e_t|/\tau_r)$, where e_t is the estimator error. The model reaches an area under the receiver operating characteristic curve (AUC) of 0.898 for detecting errors > 10 BPM. An ablation shows PPG-only features achieve an AUC of 0.898, PPG+ACC 0.888, and ACC-only 0.710, indicating that accelerometer features do not improve reliability estimation.

Coverage–error trade-off. Figure 2 shows the coverage–MAE trade-off for both the overall evaluation set and the motion-only subset. The overall curve includes static and motion windows, whereas the motion curve contains only motion windows. At full motion-window coverage, the finite-state policy achieves 9.99 BPM MAE through strategic holds. At 50% motion-window coverage, the finite-state reporting policy reaches 6.2 BPM. At 31% motion-window coverage (λ_{rej}=4), reported-window MAE reaches 2.90 BPM, comparable to static-window accuracy. Each operating point should be interpreted together with its corresponding coverage. Low-coverage operating points represent high-confidence spot measurements rather than replacements for full-coverage estimation.

Finite-state reporting policy. A learned policy chooses among three actions: accept (emit the current estimate), hold (emit the last trusted value), or reject (emit nothing). The policy state is the discretized reliability score and the number of consecutive holds.

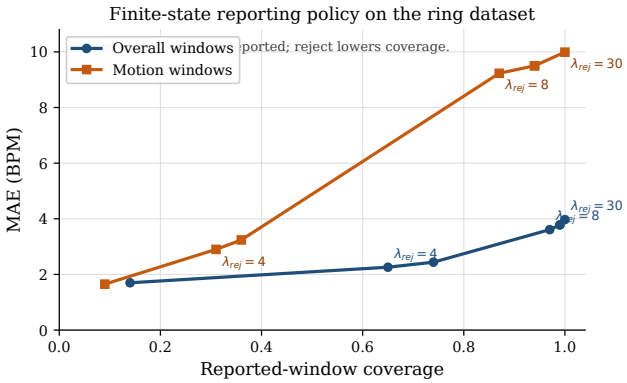


Figure 2: Coverage–MAE trade-offs for reliability-aware reporting. The overall curve includes both static and motion windows, whereas the motion curve is evaluated only on motion windows. Lower coverage corresponds to lower reported-window MAE.

Table 1: Ring-platform HR error under motion (MAE in BPM, ± standard deviation over five seeds).

Configuration	Static MAE	Motion MAE	Overall MAE
LSTM baseline	5.52 ± 0.33	14.39 ± 0.47	6.70 ± 0.27
LSTM + smoothing + ACC priors	5.33 ± 0.29	13.00 ± 0.41	6.35 ± 0.30
HGB direct-regression probe	3.85	13.08	5.08
Candidate scorer + causal Viterbi	**2.63 ± 0.06**	**10.91 ± 0.73**	**3.73 ± 0.13**

Value iteration solves for the optimal policy under a per-window cost that penalizes estimation error, stale-held values, and rejections. The rejection cost λ_{rej} controls the coverage–error trade-off. At full coverage (λ_{rej}=30), the policy emits 9.99 BPM motion MAE by holding during unreliable windows. At λ_{rej}=4, the policy reports 31% of motion windows at 2.90 BPM.

4 Evaluation

4.1 Ring-Platform Benchmark

The τ-Ring platform [14] was evaluated on 54 participants across three cohorts (stimulus-evoked, daily activities, treadmill running), totaling 49.76 h paired with clinical references (VivaLNK ECG patch, finger oximeter, respiratory band, blood pressure monitor). The motion-focused HR analysis uses 31 Ring 1 subjects under a deployable ring-only input protocol: only ring PPG, ring ACC, and ring-derived quality features are available at runtime; reference-device variables are excluded. The dataset contains 2009 windows (1742 static, 267 motion from deep-squat, striding, and shaking-head protocols).

Table 1 reports the motion-stratified results. The candidate-based estimator improves motion MAE by 3.5 BPM (24%) over the LSTM baseline and by 2.1 BPM over the ACC-prior-augmented variant. Static-window MAE reaches 2.63 BPM, approaching clinical-reference quality at rest. Motion remains the dominant error source: even the best estimator reaches 10.91 BPM on motion windows, motivating the reporting layer.

4.2 Sustained-Motion Stress Test

The sustained-motion treadmill test contains 54 subject groups, 75 sessions, and 4,491 non-overlapping 30-second windows(37.43 h), evaluated with subject-disjoint splits. The candidate HGB baseline obtains an MAE of 17.65 BPM. Adding causal Viterbi decoding reduces MAE to 15.30 BPM and raises Pearson correlation from 0.631 to 0.724. Candidate-baseline errors are heavy-tailed (window-level 90th-percentile (P90) error: 40.04 BPM; subject-median MAE: 16.05 BPM; worst-group MAE: 34.04 BPM). This auxiliary stress test further motivates temporal decoding and reliability-aware reporting.

4.3 Selective Reporting Results

Figure 2 shows the coverage–MAE trade-off. At full coverage, the finite-state policy emits 9.99 BPM motion MAE through strategic holds. At 50% motion-window coverage, the finite-state reporting policy reaches 6.2 BPM. At 31% motion coverage (λ_{rej}=4), reported-window motion MAE is 2.90 BPM—comparable to static-window accuracy. Each operating point should be interpreted together with its coverage. Low-coverage operating points represent high-confidence spot measurements from motion windows deemed reliable by the reporting model.

4.4 Real-Time ESP32 Prototype

To bridge benchmark evaluation and a physical hardware demonstration, we implement a real-time prototype around an ESP32-S3 microcontroller and an optical PPG sensor. The firmware samples PPG at 100 Hz, performs filtering and beat-event detection online, and streams synchronized raw and filtered samples to a companion browser interface. The interface displays the waveform, beat events, and current heart rate, making motion-induced corruption directly observable rather than reducing the demonstration to a static accuracy table.

The prototype establishes the sensing, timing, communication, and visualization path for migrating CITRUS to embedded hardware. The candidate scorer is compact, the causal decoder carries only past state, and the reporting policy operates window by window without future context. These properties support staged migration to ESP32-class targets while preserving the same estimation–reporting decomposition evaluated on the ring benchmark. The workshop demonstration is designed to let attendees compare rest and controlled hand motion while observing changes in the live waveform, detected beats, and displayed heart rate. These live signals provide the input path for subsequent integration of reliability-aware reporting.

4.5 Component Ablations

Table 2 summarizes controlled ablations isolating each module's contribution.

The ablations support three design conclusions for wearable deployment. First, the raw-waveform CNN encoder is unnecessary once structured candidate features are available, enabling a 3× parameter reduction. Second, accelerometer-guided candidate selection reduces motion MAE by 1.0 BPM. Third, causal decoding performs comparably to full-sequence decoding (10.66 vs. 10.88 BPM) while requiring no future windows, making it suitable for online inference.

Table 2: Controlled component ablations supporting efficient causal deployment on the ring motion subset. Evaluation settings are specified in the corresponding text.

Module	Ablation	Key result
Candidate scoring	Remove raw-waveform CNN encoder	Δoverall MAE −0.01 BPM; 85k → 28k parameters
Candidate generation	Remove ACC segment selection	+1.0 BPM motion MAE
Temporal decoder	Causal (online) vs. full-sequence	10.66 vs. 10.88 BPM on a fixed run; no future windows

4.6 Deployment Footprint

We deploy and benchmark the CITRUS feature-vector inference core—the candidate HGB scorer, causal Viterbi decoder, and reliability HGB—on an ESP32-S3 revision 0.2 at 160 MHz. The compact tree ensembles occupy 145,280 B; the benchmark firmware occupies 447,856 B, including embedded golden test vectors. Across 60 warm window inferences (five passes over a 12-window sequence), mean, median, 95th-percentile (p95), and maximum latencies are 156.429, 188.470, 197.943, and 197.949 ms, respectively, with 392,324 B minimum free heap. Candidate and Viterbi output selections match the desktop implementation on all 12 golden windows, and the maximum absolute reliability-output difference is 3.815×10^{-6}. Applying the ESP32-S3 datasheet range for single-core active current at 160 MHz gives an estimated inference energy of 20.60–28.19 mJ per window at the mean latency and 26.06–35.67 mJ per window at p95. The benchmark begins from precomputed candidate features and excludes raw-signal filtering and candidate generation; its latency is therefore an inference-core measurement rather than end-to-end sensor latency.

5 Discussion and Conclusion

CITRUS separates estimation from reporting for motion-robust wearable HR monitoring. On the 31-subject motion subset, candidate-based causal decoding reduces motion-window MAE from 14.39 to 10.91 BPM, while reliability-aware reporting reaches 6.2 BPM at 50% motion-window coverage. On the separate sustained-motion treadmill test, causal Viterbi reduces MAE from 17.65 to 15.30 BPM and improves Pearson correlation from 0.631 to 0.724. These results show that temporal decoding improves full-coverage estimation, whereas selective reporting provides high-confidence spot measurements when lower coverage is acceptable.

The feature-vector inference core runs on an ESP32-S3 in an average of 156.429 ms per window using 145,280 B of model storage. Its datasheet-based compute-energy estimate is 20.60–28.19 mJ per window at the mean latency. The live 100-Hz sensing and inference benchmarks currently use separate firmware paths; raw-signal candidate generation remains to be integrated. Sustained motion can make every candidate unreliable, and the transition penalty may over-smooth rapid HR changes. Future work will complete end-to-end hardware integration and evaluate motion-adaptive transitions in broader populations.

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